

Corrosion RC Beam Optimizer — Scientific Report

PhD Research — Corrosion RC Beam Optimization

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This report presents the complete results of the Corrosion RC Beam Optimizer research pipeline. The study applies a Neural Network-NSGA-III Genetic Algorithm framework to predict the residual flexural capacity $R(\%)$ of corroded reinforced concrete beams, benchmarked against ACI 318-19 and the state-of-the-art ML literature (Zhang et al., 2025). Two hierarchical performance layers must be surpassed: L1: $R^2 > 0.9$ (ACI threshold), and L2: $R^2 > 0.972$ (SOTA threshold). Statistical significance is confirmed via Wilcoxon, Bootstrap CI, 10-Fold CV, Cohen's d, and McNemar tests.

NSGA-III Configuration

Parameter	Value
Population Size	40
Max Generations	80
Max Restarts	3
Consistency Window	15
Crossover Rate	0.85
Mutation Rate	0.15
Fitness W1 (R^2)	0.6
Fitness W2 (ACI improve)	0.25
Fitness W3 (penalty)	0.15
L1 Target R^2	0.9
L2 Target R^2	0.972

ACI 318-19 Benchmark Results

Metric	Value
R2	0.8839
Rmse	8.2289
Mae	5.0617
Mape	29.9
Ratio Mean	1.1845
Ratio Std	0.5874
Ratio Min	0.2735
Ratio Max	5.2832
Underestimate Pct	48.5
N Specimens	804

MLP Baseline — Training & Test Metrics

Metric	Value
Train.Split	Train
Train.R2	0.9882
Train.Rmse	2.5836
Train.Mae	1.3976
Train.Mape	9.76
Train.L1 Broken	True
Train.L2 Broken	True
Train.Timestamp	2026-04-07 07:24:35.866131
Test.Split	Test
Test.R2	0.9695
Test.Rmse	4.4713
Test.Mae	2.342
Test.Mape	13.18
Test.L1 Broken	True
Test.L2 Broken	False
Test.Timestamp	2026-04-07 07:24:35.869782
Cv.Cv R2 Mean	0.9671
Cv.Cv R2 Std	0.0097
Cv.Cv Rmse Mean	0.166

Cv.Cv Rmse Std	0.0207
Cv.N Folds	10
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SHAP Feature Importance

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PySR Discovered Equation

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Statistical Validation Results

Metric	Value
Wilcoxon.Test	Wilcoxon Signed-Rank (one-sided, less)
Wilcoxon.Statistic	1279.0
Wilcoxon.P Value	0.0
Wilcoxon.Alpha	0.05
Wilcoxon.Significant	True
Wilcoxon.Interpretation	Model errors are statistically SMALLER than ACI errors (p=0.0000 < α=0.05). The improvement over ACI 318-19 is statistically significant.
Wilcoxon.Median Err Model	0.7484
Wilcoxon.Median Err Aci	2.7885
Bootstrap.N Bootstrap	1000
Bootstrap.Ci Level	0.95
Bootstrap.R2	0.987
Bootstrap.R2 Ci	[0.9791, 0.9917]
Bootstrap.Rmse	2.921
Bootstrap.Rmse Ci	[2.3059, 3.4684]
Bootstrap.Mae	1.7452
Bootstrap.Mae Ci	[1.3973, 2.1255]
Bootstrap.R2 Boot Mean	0.9865
Bootstrap.R2 Boot Std	0.0032
Kfold.N Folds	10
Kfold.R2 Mean	0.9671
Kfold.R2 Std	0.0097
Kfold.R2 Min	0.9501
Kfold.R2 Max	0.9819
Kfold.Rmse Mean	0.166
Kfold.Rmse Std	0.0207
Kfold.Mae Mean	0.0956
Kfold.Mae Std	0.0088
Kfold.L1 All Folds	True
Kfold.L2 All Folds	False
Cohens D.Cohens D	0.5814
Cohens D.Magnitude	medium

{'fold': 4, 'R2': 0.9582, 'RMSE': 0.2021, 'MAE': 0.1019}, {'fold': 5, 'R2': 0.9724, 'RMSE': 0.1825, 'MAE': 0.101}, {'fold': 6, 'R2': 0.9811, 'RMSE': 0.1338, 'MAE': 0.0956}

Cohens D.Mean Diff	3.3458
Cohens D.Std Diff	5.7545
McNemar.Test	McNemar (exact, continuity corrected)
McNemar.Tolerance Pct	10.0
McNemar.B Model Ok	23
McNemar.C Aci Ok	0
McNemar.Chi2 Statistic	21.0435
McNemar.P Value	4e-06
McNemar.Significant	True
McNemar.Model Accuracy	99.38
McNemar.Aci Accuracy	85.09
Summary.Wilcoxon Significant	True
Summary.Bootstrap R2 Lower Bound	True
Summary.Kfold L1 All Folds	True
Summary.Kfold L2 All Folds	False
Summary.Effect Size Large	False
Summary.McNemar Significant	True
Summary.Model R2	0.987
Verdict	■ STATISTICAL VALIDATION PASSED — Benchmark improvement confirmed with p
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Figures Gallery

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Conclusion & Decision

■ STATISTICAL VALIDATION PASSED — Benchmark improvement confirmed with $p < 0.05$.

Test	Result
Wilcoxon Significant	✓
Bootstrap R2 Lower Bound	✓
Kfold L1 All Folds	✓
Kfold L2 All Folds	✗
Effect Size Large	✗
McNemar Significant	✓

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